

# VEDANG TRIVEDI

Software Engineer | Data Engineer | Distributed Systems, Python & ML Systems

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 [github.com/VEDANG2024](https://github.com/VEDANG2024)  Ahmedabad, Gujarat, India

## PROFILE SUMMARY

Software Engineer specializing in Conversational AI, LLMs, LangChain, and distributed software engineering. Designed, built, and shipped enterprise applications at Kraft Heinz used daily by **50+ people** across **10 towers**, leading technical delivery and automation estimated to save **1,600+ hours** a year. Technical lead on AI systems research, engineering an entropy-aware LLM-routing model that cuts inference cost by **78%**, alongside query-plan optimization models.

## TECHNICAL SKILLS

**Languages & APIs:** Python, JavaScript, SQL, REST APIs, JSON/YAML, C, C++

**AI & ML Systems:** Conversational AI, LLMs, Agentic AI, LangChain, NLU (NLTK/spaCy), Prompt Engineering, QLoRA/PEFT, XGBoost

**Databases:** MongoDB (NoSQL), Redis (NoSQL), PostgreSQL, MySQL

**DevOps & Tools:** Git Workflows, CI/CD, Technical Documentation, Power Apps, Power Automate, Figma

**Data & Analytics:** Power BI, Pandas/NumPy

## PROFESSIONAL EXPERIENCE

### Kraft Heinz

*KHMS/AIT Intern, GCC India*

**Feb 2026 – Present**

*Ahmedabad, Gujarat*

- Engineered the AIT Power App end-to-end (5 screens, 20 SharePoint columns across 4 data types) after feasibility checks and process mapping, deploying a standardized 7-stage lifecycle model to **50+ users** across **10 towers**.
- Implemented 6+ conditional logic rules (field visibility, record locking, cascading filters, form validation) and 2 automated notification triggers/webhooks, cutting an estimated **1,130+ hours a year** of manual status-update and reporting effort.
- Debugged and resolved **12 documented defects** through root-cause analysis during User Acceptance Testing (UAT) and architecture audits, maintaining technical documentation and shipping with zero external budget.
- Migrated Health Check data onto a governed SharePoint backend (**5 lists**) and engineered the Health Check and OpEx Survey app suite – wizard-based flow, dependent dropdowns, submission validation – for **50+ users** across **11 towers**, cutting submission errors to near-zero and saving an estimated **267 hours a year**. Enforced role-based access control to maintain platform data governance.

### Karma Technolabs

*Web Development Intern*

**Jan 2024 – Apr 2024**

*Ahmedabad, Gujarat*

- Enhanced search, filter, and upload API functionality on a live CRM platform, contributing end-to-end across Git workflows, testing, and production deployment.

## PROJECTS

### UGAD-Lite: Uncertainty-Guided Adaptive Distillation | *ML Systems*



- Engineered an agentic entropy-conformal router that dynamically directs user prompts between a large LLM (GPT-4o) and a fine-tuned Small Language Model (Phi-3-Mini via QLoRA/NLU), applying prompt engineering and conformal prediction for statistical safety guardrails.
- Cut LLM inference cost and token usage by 78% at a 94.8% task success rate on GSM8K and synthetic arithmetic benchmarks, with a <5% false-negative rate.

### CardEst | *SQL, Query Optimization*



- Built an XGBoost-based cardinality estimator trained on the TPC-H dataset to improve query-optimizer execution-plan accuracy and database performance.
- Cut query latency and resource usage in 70% of evaluated cases by improving Pearson correlation and reducing median estimation error.

### Raft Census (DBFS) | *Distributed Systems, Raft Consensus*



- Built a language-agnostic, decentralized file-storage system using Raft cluster consensus (Redis + MongoDB NoSQL backend), with work-stealing task distribution to support concurrent multi-client operations with efficient search and caching.

### UML2PlantUML | *LLMs, Machine Learning*



- Built a small-scale, local reproduction of the approach from Sadeghi et al. (MODELS 2026, arXiv:2607.28987) for handwritten UML-to-PlantUML generation, preprocessing datasets and authoring technical documentation.
- Fine-tuned and benchmarked 7 vision-language models (ViT-BART, BLIP, Flan-T5, ViT-GPT2, Pix2Struct, CLIP, MiniCPM) to generate PlantUML code from class diagram images, establishing evaluation pipelines and deploying the trained model on Kaggle.

### Distributed Database System | *2PC, Query Optimization*



- Engineered a distributed database with a coordinator-worker topology, implementing horizontal/vertical data fragmentation and a global data dictionary for location-transparent query routing.
- Implemented Two-Phase Commit (2PC) for multi-node transaction atomicity, JSON payloads, and semi-join query optimization, cutting cross-node bandwidth by 60–70% and query latency by 3–4x versus centralized execution.

## EDUCATION

### DA-IICT (Dhirubhai Ambani Institute of Information and Communication Technology)

*M.Tech, ICT (Software Systems) | CPI: 7.47/10*

**2024 – 2026**

*Gandhinagar, Gujarat*

### Gujarat Technological University

*B.E., Computer Engineering | CGPA: 7.77/10*

**2020 – 2024**

*Ahmedabad, Gujarat*

## POSITIONS OF RESPONSIBILITY

### Teaching Assistant – Software Engineering & Object-Oriented Programming

*DA-IICT, Gandhinagar*

**Aug 2024 – Present**

*Gandhinagar, Gujarat*

- Provided leadership and mentoring in labs for 100+ students, evaluated code quality and assignments, and guided technical documentation for Software Engineering and Object-Oriented Programming.

ACHIEVEMENTS

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- GATE CS 2024 qualifier.